Optimizers and their convergence

Consider O: a simplified Adam-style optimizer without weight decay that has iterates

$\theta t+1 \leftarrow \theta t - \alpha t M t \nabla f t(\theta t)$ (4)

where ft

is the loss at iteration t and αt

is the step size (learning rate).

Further suppose that the adaptive scaling matrix Mt

is recomputed over each epoch of training and just

consists of a diagonal populated by the inverses of the square roots of the mean squared value for the

gradients during the epoch for that specific coordinate.

For this part, we have exactly n = 1 training point corresponding to the single equation

$[1, 0.1, 0.01]\theta = 1$ (5)

with a 3-dimensional learnable parameters θ. Suppose that we start with $\theta 0 = 0$ and use squared loss

$ft(\theta) = (1 - [1, 0.1, 0.01]\theta)$

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(a) What specific vector θ would standard vanilla SGD (i.e. (4) with Mt = I and $\alpha t = \alpha$) converge

to assuming $\alpha > 0$ was small enough to give convergence?

(b) What specific vector θ would the simplified version of Adam O converge to assuming appropriate

step-sizes $\alpha t > 0$ to give convergence?

(c) Consider a learning approach that first did training input feature rescaling (so that each feature had

unit second-moment), then ran SGD to convergence, and then converted the solution for the rescaled

problem back to the original units. What specific vector θ would it give as

its final solution (for use
in original coordinates)?

Thought for 1m 39s

(a) Standard vanilla SGD converges to the minimum-norm solution satisfying the equation, which is the vector
θ = \left[\frac{10000}{10101}, \frac{1000}{10101}, \frac{100}{10101} \right].

To arrive at this solution, note that the problem is an underdetermined linear system [1, 0.1, 0.01]θ = 1 minimized via squared loss starting from $\theta_0 = 0$. Vanilla SGD (gradient descent) converges to the pseudoinverse solution $\theta = a^T / ||a||^2$ where a = [1, 0.1, 0.01]. Compute $||a||^2 = 1 + 0.01 + 0.0001 = 1.0101 = 10101/10000$, so θ = a × (10000/10101).

(b) The simplified Adam-style optimizer converges to the vector
θ = \left[\frac{100}{111}, \frac{100}{111}, \frac{100}{111} \right].

To arrive at this solution, observe that the update simplifies to adding α_t to each component of θ_t due to the adaptive scaling canceling the differences in gradient magnitudes (which are proportional to a_i). The trajectory is θ = c[1, 1, 1] where c satisfies [1, 0.1, 0.01] · (c[1, 1, 1]) = 1, so c(1 + 0.1 + 0.01) = 1 yielding c = 1/1.11 = 100/111.

(c) This approach converges to the vector
θ = \left[\frac{1}{3}, \frac{10}{3}, \frac{100}{3} \right].

To arrive at this solution, rescale features to unit second-moment: since the second moment of each feature is a_i^2, divide by a_i to get a' = [1, 1, 1]. Run SGD on the rescaled problem, converging to θ' = [1/3, 1/3, 1/3]. Convert back: $\theta_j = \theta'_j / a_j$, yielding the above vector.

Please rerender the latex

Here is the re-rendered LaTeX for the provided mathematical expressions:

(a) The vector θ for standard vanilla SGD is:

$$\theta = \left[\frac{10000}{10101}, \frac{1000}{10101}, \frac{100}{10101} \right]$$

(b) The vector θ for the simplified Adam-style optimizer is:

$$\theta = \left[\frac{100}{111}, \frac{100}{111}, \frac{100}{111}\right]$$

(c) The vector θ for the feature rescaling approach is:

$$\theta = \left[\frac{1}{3}, \frac{10}{3}, \frac{100}{3}\right]$$

Please also rerender the explanations

Below, I've re-rendered the explanations for each part with LaTeX integrated where appropriate, maintaining clarity and conciseness while ensuring the mathematical expressions are properly formatted.

(a) **Standard Vanilla SGD Convergence**

Vanilla SGD converges to the minimum-norm solution satisfying the equation $[1, 0.1, 0.01]\theta = 1$, which is the vector:

$$\theta = \left[\frac{10000}{10101}, \frac{1000}{10101}, \frac{100}{10101}\right].$$

This result comes from solving the underdetermined linear system with the squared loss $f_t(\theta) = (1 - [1, 0.1, 0.01]\theta)^2$, starting from $\theta_0 = 0$. Vanilla SGD (equivalent to gradient descent with $M_t = I$) converges to the pseudoinverse solution $\theta = a^T/\|a\|^2$, where $a = [1, 0.1, 0.01]$. Compute the norm: $\|a\|^2 = 1 + 0.01 + 0.0001 = 1.0101 = \frac{10101}{10000}$. Thus, $\theta = a \cdot \frac{10000}{10101}$, yielding the above vector.

(b) **Simplified Adam-style Optimizer Convergence**

The simplified Adam-style optimizer converges to the vector:

$$\theta = \left[\frac{100}{111}, \frac{100}{111}, \frac{100}{111}\right].$$

For this optimizer, the adaptive scaling matrix M_t normalizes the gradients by the inverse square roots of their mean squared values over the epoch. Since there's one training point, the gradient is $-2(1 - [1, 0.1, 0.01]\theta)[1, 0.1, 0.01]$, and the scaling makes the update direction proportional to $[1, 1, 1]$. The update effectively adds α_t to each component of θ_t. The solution lies along $\theta = c[1, 1, 1]$, where c satisfies $[1, 0.1, 0.01] \cdot c[1, 1, 1] = 1$, so $c(1 + 0.1 + 0.01) = 1$, giving $c = \frac{1}{1.11} = \frac{100}{111}$.

(c) **Feature Rescaling with SGD Convergence**

The learning approach with feature rescaling converges to the vector:

$$\theta = \left[\frac{1}{3}, \frac{10}{3}, \frac{100}{3}\right].$$

First, rescale the features to have unit second-moment. The second moment of each feature is a_i^2, where $a = [1, 0.1, 0.01]$. Dividing by a_i, the rescaled features are $a' = [1, 1, 1]$. Running SGD on the rescaled problem $[1, 1, 1]\theta' = 1$ with squared loss yields the minimum-norm solution $\theta' = \left[\frac{1}{3}, \frac{1}{3}, \frac{1}{3}\right]$. Converting back to original coordinates, $\theta_j = \theta'_j / a_j$, so $\theta = \left[\frac{1/3}{1}, \frac{1/3}{0.1}, \frac{1/3}{0.01}\right] = \left[\frac{1}{3}, \frac{10}{3}, \frac{100}{3}\right]$.